Java 21 Virtual Threads & Structured Concurrency

Java 21.0.6
VERIFIED RUNTIME

KEY SYNTAX & IDIOMS

Virtual Thread Creation
```
Thread.startVirtualThread(() -> { ... });
```
Spawns a lightweight daemon thread

ExecutorService Idiom
```
try (var exec =
Executors.newVirtualThreadPerTaskExecutor())
```
Auto-closes tasks with StructuredTaskScope

Record Syntax
```
public record User(Long id, String name) {}
```
Immutable transparent data carrier

Pattern Matching
```
case String s when s.length() > 5 -> ...
```
Type switch with guard conditions

COMMON PITFALLS & FIXES

Pitfall: Synchronized blocks pin carrier threads
Rule: Use ReentrantLock instead of synchronized blocks in virtual threads

Pitfall: ThreadLocal pooling overhead
Rule: Avoid pooling virtual threads; create anew per request

Pitfall: Missing -Xlint:removal flag
Rule: Build with -Xlint:deprecation,removal in compiler arguments

TINY VERIFIED EXAMPLE: JAVA 21 THREAD RUNNER

SOURCE CODE:
```
public class Solution {
public static void main(String[] args) throws Exception {
var t = Thread.startVirtualThread(() -> System.out.println("Virtual!"));
t.join();
}
}
```

REAL CAPTURED OUTPUT:
```
Virtual!
```